

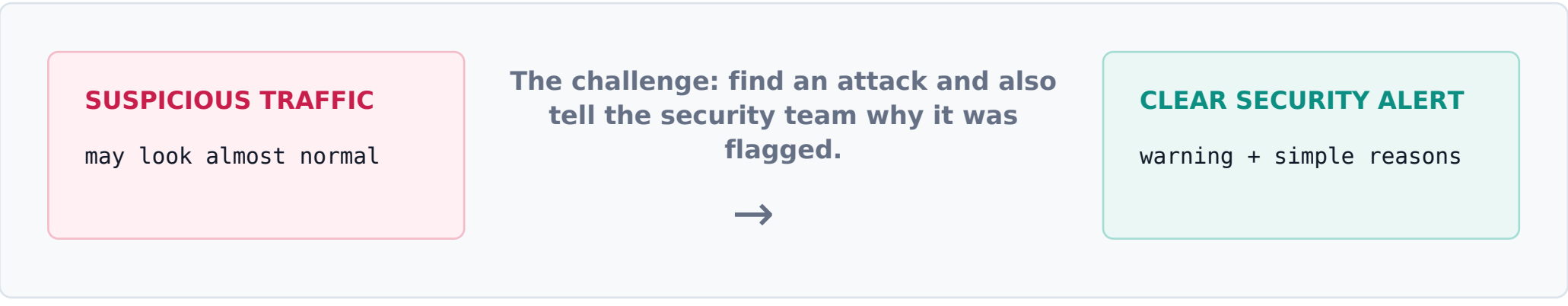
Enhanced-IDS

A smart and explainable system that learns the difference between normal and suspicious network traffic.

Alamro et al. | Scientific Reports 2025, 15:33848 | DOI: 10.1038/s41598-025-06761-9

What exactly is this work about?

This paper builds a smart security guard for computer networks. It studies each network connection, decides whether it looks Normal or Suspicious, and shows which information influenced the decision. The complete research model is named EIDCDR-XAIADL.



WHAT THE SYSTEM RECEIVES

INFORMATION RECEIVED

Details about each network connection

For example: connection time, service used, amount of data transferred and connection status.

DATASET 1

NSL-KDD

50,000 examples: 25,000 normal connections and 25,000 suspicious connections; 42 details per connection.

DATASET 2

CICIDS 2017

50,000 examples: 25,000 normal connections and 25,000 suspicious connections; 78 details per connection.

Testing used in the paper. The system learns to choose between two classes: normal traffic and suspicious traffic. Its performance is checked during training from 500 to 3000 rounds.

HOW THE SYSTEM WORKS

1 Put values on one scale

Mean normalization makes differently sized values easier for the model to compare.

2 Keep useful information

MVO removes less useful details and keeps the ones that best separate normal and suspicious traffic.

3 Find small warning patterns

A CNN learns combinations of connection details that often appear in attacks.

4 Understand order and focus

BiGRU studies how information is connected, while attention focuses on the strongest warning signs.

5 Choose the best settings

ALO automatically tries different model settings and keeps the combination with fewer mistakes.

6 Explain the result

SHAP shows which connection details pushed the model toward normal or suspicious.

1 Connection

2 Prepare

3 Choose

4 Decide

5 Explain

HOW ONE ALERT IS CREATED

Look for warning signs

The CNN searches the connection details for combinations commonly linked with attacks.

Understand the full picture

BiGRU connects related information, and attention gives extra importance to the strongest signs.

Give a simple reason

SHAP points out the details that increased or decreased the suspicious-traffic score.

A useful way to read one alert

1 The system marks a network connection as Suspicious.

2 It lists the connection details that most influenced the warning.

3 A security analyst checks the warning and decides whether action is needed.

WHAT THE SYSTEM GIVES

DECISION

Normal or Suspicious

The result given for each network connection.

CONFIDENCE

How sure is the system?

A score that helps security teams examine the strongest alerts first.

EXPLANATION

Why was it flagged?

SHAP identifies the connection details that influenced the result.

NEXT STEP

Analyst review

A security analyst checks important warnings and decides what to do.

Practical purpose. The system acts as an early warning tool. It helps a security analyst find unusual connections faster; it does not replace the analyst's final decision.

RESULTS REPORTED IN THE PAPER

Results on two datasets containing equal numbers of normal and suspicious examples, after 3000 training rounds.

DATASET	FEATURES	ACCURACY	PRECISION	RECALL	F1
NSL-KDD	42 → 24	99.19%	99.19%	99.19%	99.19%
CICIDS 2017	78 → 46	99.12%	99.12%	99.12%	99.12%

42 → 24

NSL-KDD details kept
18 less-useful details removed

78 → 46

CICIDS 2017 details kept
32 less-useful details removed

8.78 s

reported processing time
on NSL-KDD

10.52 s

reported processing time
on CICIDS 2017

Accuracy improves during training

Dataset	500	1000	1500	2000	2500	3000
NSL-KDD	99.1%	99.15%	99.17%	99.18%	99.19%	99.19%
CICIDS 2017	99.0%	99.05%	99.08%	99.10%	99.11%	99.12%

Read the scores carefully

Both datasets contain equal numbers of normal and suspicious examples, which is easier than many real networks. The results come from the paper and were not independently repeated. The paper does not show testing on live network traffic.

WHERE IT CAN BE USED

- Security alert screening. Place the most suspicious connections at the top of the review list.
- Company network monitoring. Continuously separate common traffic from connections that need attention.
- Understanding AI decisions. Show the network details that caused each warning instead of giving only a label.

What the results actually show. The system performed very well on the two prepared datasets and used fewer connection details. Before real use, it must also be tested with rare attacks, new attack types, future traffic and a live network.

CHECKS NEEDED BEFORE REAL USE

New attack types

Check whether it can detect attacks that were not shown during training.

Realistic traffic mix

Test where normal connections are common and attacks are rare.

Future traffic

Train with older traffic and test with data collected later.

Speed and workload

Measure how fast it works and how many alerts people must review.

Alamro et al., 'Enhanced intrusion detection in cybersecurity through dimensionality reduction and explainable artificial intelligence,' Scientific Reports 15 (2025), 33848. DOI: 10.1038/s41598-025-06761-9. Figures and values follow the paper; results are author-reported, not independently reproduced.